

# Practice II

Polynomial regression

# Specifications

- The dataset `cal_housing.csv` will be used in this practice
  - Columns *longitude*, *latitude*, *housingMedianAge*, *totalRooms*, *totalBedrooms*, *population*, *households*, *medianIncome* are features of houses
  - Column *medianHouseValue* is the target
- Using the dataset do the following:
  - Load the dataset
  - Create a train set with 80% of data and 20% for test (set *random\_state* = 0)
  - Create a validation set with ten folds using the train set

# Specifications

- Using the validation (80% of data in 10 folds) set do the following for each fold:
  - Using data with no scalation
    - Use stochastic gradient descent to create a line that best fit to data. Parameters *max\_iter*, *learning\_rate* and *eta0* should be adjusted. Report MSE and coefficient of determination  $r^2$
    - Use polynomial regression with stochastic gradient descent to create a curved line that best fits the data. Use polynomials of degree 2 and 3 and adjust the parameters *max\_iter*, *learning\_rate* and *eta0*. Report MSE and coefficient of determination  $r^2$
    - Store the MSE and  $r^2$  values for each fold and report the mean of the 10 folds for each experiment
  - Using the data with *standard scaler* repeat all the previous experiments
  - Using the data with *robust scaler* repeat all the previous experiments

# Specifications

- Using the train set (80% of data), train a model with the type of regression (linear or polynomial), parameters and scaling method (none, standard or robust) that obtained the best average performance in the validation set
- Using the above model perform the prediction of the test set data and report MSE and  $r^2$

# Evidence

- Source code
- A document with the following information:
  - An introduction explaining the problem to be solved and a description of the linear regression method used
  - Table reporting the results on the validation set and some conclusions that can be drawn from them, indicating the characteristics (regression type, scaling, etc.) of the model selected for the test set.

| Type of regression | Scaling method | Learning rate | Number of iterations | Average MSE | Average $r^2$ |
|--------------------|----------------|---------------|----------------------|-------------|---------------|
|                    |                |               |                      |             |               |

- MSE and  $r^2$  obtained in the test set
- General conclusions of the practice, mentioning the difficulties encountered and any ideas to improve the results obtained.